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Andrea *Bonacci*

AI ENGINEER — LLM & AGENTIC SYSTEMS

PROFILE

From Data Scientist to **AI Engineer**, *with* **over 4 years of experience** *in data-driven consulting for enterprise clients. I design and ship production AI systems: agents and agentic workflows with MCP and tool use, LLM integration (OpenAI, Anthropic), RAG and real-time voice AI. Solid background in end-to-end data analysis: pipelines, BI dashboards and NLP models, from ETL to deployment with FastAPI and Docker.* **Academic tutor** *for university interns and responsible for* **technical interviews** *for post-acquisition hiring. I also delivered an intensive* **8-hour university course** *on AI applied to coding.*

PROFESSIONAL EXPERIENCE

LINKALAB S.R.L. · CAGLIARI — part of Lynx Group since 2025 — Nov 2021 → present

Consulting company — Internal growth

AI Engineer & Data Analyst

Oct 2025 → present

AI Engineering focus: AI agents and agentic workflows with MCP, tool use, sub-agents and LLM integration (OpenAI, Anthropic). Production AI systems on client projects, including real-time conversational voice AI, agents for document and accounting automation, and RAG for compliance validation against regulatory corpora. Claude Code, Windsurf and Gemini CLI in production. Academic tutor for university interns. Technical interviews for hiring Data Scientists and AI Engineers post-acquisition.

`AI Agents` `MCP` `LLM Integration` `Realtime Voice AI` `RAG` `Document AI` `Claude Code` `Academic Tutor` `Technical Hiring`

Data Analyst & Data Scientist

Jul 2022 → Oct 2025

End-to-end data pipelines and BI dashboards (Power BI, QuickSight, Tableau, Looker Studio) for enterprise clients. NLP, sentiment analysis, time series. Deployment with FastAPI and Docker on AWS.

`Python` `SQL` `Power BI` `QuickSight` `Tableau` `NLP` `FastAPI` `Docker` `AWS S3/Athena`

Data Scientist — contract

Apr 2022 → Jun 2022

Post-internship hire. First autonomous data analysis and modeling projects for enterprise clients.

Data Scientist — curricular internship

Nov 2021 → Mar 2022

Master's thesis internship. Development of an Italian Emotion Analysis system for online reputation monitoring.

AT A GLANCE

4+	18+
years of professional experience	open source projects

DIFFERENTIATORS

AI + BUSINESS

Business degree + Data Science master's = I build AI that solves real problems, from idea to deployment.

AI ENGINEERING

Advanced use of LLMs, AI agents and MCP to build intelligent workflows and automate complex tasks.

UNIVERSITY SPEAKER

AI applied to coding course, University of Cagliari, April 2026 — 8 hours.

MENTORING & HIRING

Academic tutor for university interns. Leading technical interviews for post-acquisition team expansion.

SKILLS

AI & LLM

AI Agents · **MCP** · **Prompt Eng.** · **RAG** · OpenAI API · Anthropic API · Tool Use · Function Calling · Document AI · Human-in-the-Loop · Realtime Voice · Sub-agents · Hugging Face · Ollama · LLM Eval · Agentic Coding Config

DATA SCIENCE

Python · **SQL** · **NLP** · **Sentiment Analysis** · Pandas · Time Series · Computer Vision

BUSINESS INTELLIGENCE

Power BI · QuickSight · Tableau · Looker Studio

Speaker — University Course

April 2026

UNIVERSITY OF CAGLIARI

Design and delivery of two intensive sessions (**8 hours total**) on *"Artificial Intelligence Applied to Coding"*. LLM architecture, advanced prompt engineering, agentic coding with CLI tools, Model Context Protocol (MCP) integration and custom Skills creation for AI agents. Hands-on workshops where students built their first AI-assisted development workflows.

`AI for Coding` `Prompt Engineering` `LLM` `MCP` `University Lecture`

CLIENT PROJECTS

ON A CONTRACT BASIS FOR AN ENTERPRISE CLIENT · DETAILS ANONYMIZED FOR CONFIDENTIALITY

Real-Time Conversational Voice AI Systems

Low-latency, voice-to-voice AI conversation systems with automatic evaluation of the interactions.

- Integration of leading real-time voice AI providers (OpenAI Realtime API, Gemini Live) and streaming transcription (AssemblyAI).
- Multi-provider architectures with fallback and circuit breaker for production resilience.
- RAG knowledge bases (embeddings, vector search) queried by the AI via function calling.
- Automatic evaluation with anti-hallucination validation and multilingual prompt engineering, integrable into heterogeneous production stacks.

`Realtime Voice AI` `OpenAI Realtime API` `RAG` `Function Calling` `Circuit Breaker` `pgvector`

AI Agents for Document & Finance Automation

AI agents for document and accounting process automation, such as automatic reconciliation of related documents.

- LLM agents with native function calling and validated structured output (Pydantic), with advanced prompt engineering.
- Document AI to extract structured data from heterogeneous documents (XML, PDF, digitally signed files) via parsing and information extraction.
- Deterministic matching engines with rules, tolerances and confidence scoring.
- Integration with business and ERP systems, plus human-in-the-loop flows for reviewing uncertain cases.

`AI Agent` `Function Calling` `Document AI` `Information Extraction` `Fuzzy Matching` `ERP Integration`

RAG & Document Intelligence for Compliance

AI systems that validate document compliance (contracts, policies, playbooks) against a regulatory corpus, with source citation and human review.

- RAG pipelines with hybrid search (BM25 and semantic embeddings) and domain-specific chunking.
- Multi-phase agentic orchestration over LLMs for issue generation and structured verdicts with regulatory references.
- Reliability through validated output and anti-hallucination via mandatory back-links to sources.
- Human-in-the-loop flows with checkpoint, resume and audit trail; asynchronous FastAPI backend with WebSocket updates.

`RAG` `Hybrid Search` `LLM Orchestration` `Structured Output` `Human-in-the-Loop` `FastAPI`

MLOPS & CLOUD

Docker · FastAPI · AWS S3 · Athena · AWS Bedrock · pgvector · Flask · GitHub

LANGUAGES

Italian	Native
English	Fluent

PERSONAL PROJECTS & OPEN SOURCE

GITHUB.COM/ANDREABONN · 20 PUBLIC REPOSITORIES

AI & Automation

Ongoing

Two GitHub Actions on Marketplace with multi-provider LLM architecture and fallback: AI PR Reviewer (code review, key rotation) and AI Changelog Generator (release notes, self-evaluation, multilingual). Video anonymization with YOLO/CV, Chrome extension for LLM-powered article/PDF summaries, multilingual TTS.

`Python` `GitHub Actions` `GitHub Marketplace` `YOLO` `Computer Vision` `LLM API` `Chrome Extension`

AI-Powered Web Apps

Ongoing

BetTracker — sports betting tracker with screenshot OCR, automatic result resolution, AI chat and advanced statistics. PWA with offline, OAuth and cloud sync.

`React` `TypeScript` `Gemini AI` `Firebase` `PWA` `Tailwind CSS`

Full-Stack Web Apps

Ongoing

Production-grade PWAs with end-to-end ownership: Meteo & Spiagge — multiple Open-Meteo APIs, 0-10 beach scoring, Leaflet maps, Recharts, Firestore community layer, offline; PlotMark — movie/TV tracking with TMDB, statistics, year-in-review and spoiler-safe social. Plus finance, health, shared expenses. End-to-end ownership.

`React` `TypeScript` `JavaScript` `PWA` `Leaflet` `Firebase` `Cloud Sync`

Developer Tools & CLI

Ongoing

Claude Code Tools Explorer — TUI for config inspection (Textual/Rich). Command Quiver — desktop prompt & shell manager (GTK4, SQLite). Telegram bot for remote multi-PC shell (zero infrastructure, heartbeat, audit log, systemd), file manager extensions, clipboard image CLI, audio renaming via AcoustID, and smart clock-in notifications.

`Python` `JavaScript` `asyncio` `Textual/Rich` `Telegram Bot` `Shell` `Linux` `systemd` `CLI` `Chrome Extension`

Security & Responsible Disclosure

Aug 2026

Reported a critical RCE vulnerability (unauthenticated code execution as SYSTEM) in WinBoat, an open source project. Private responsible disclosure with runtime-verified patches, fixed within 4 hours of first contact. Credited in the v0.9.1 security release.

`Responsible Disclosure` `RCE` `Open Source`

EDUCATION

Master's Degree — Data Science & Business Informatics

2022

University of Pisa

Thesis: Italian Emotion Analysis for online reputation monitoring

Bachelor's Degree — Business Administration

2018

University of Pisa